Rehan Saqib

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Education

National University of Science and Technology (NUST) — Islamabad, Pakistan
Bachelor of Engineering in Computer Engineering — *Sept. 2024 – Sept. 2028*

Experience

RiseTech — Islamabad, Pakistan
Software Engineering Intern — *June 2026 – Aug. 2026*

- Developed a full-stack FYP evaluation portal with **Next.js, Firebase, and MongoDB**, implementing **JWT authentication**, role-based admin and faculty workflows, CSV import/export, and end-to-end evaluation progress tracking used throughout the university's FYP evaluation cycle.
- Containerized the application with **Docker** and deployed it to **Google Cloud Platform** using **Cloud Build**, automating build-and-deploy workflows for consistent releases; integrated **Firebase** real-time listeners for authentication and live evaluation status updates.

Projects

Flotilla – Distributed Consensus Engine (Raft from Scratch) | *Go, React* — GitHub | Live Demo

- Implemented the **Raft consensus algorithm from scratch** in Go (no consensus library) across a 10-node distributed key-value store to reason directly about consistency under **leader failures, network timing differences, and concurrent operations**.
- Built core Raft mechanics – **leader election, log replication, and commit-index advancement** – then extended it with **log compaction via snapshotting** and **linearizable ReadIndex reads**, guaranteeing committed data is never silently lost or served stale.
- Designed each node as an independent process communicating over HTTP/JSON RPC, coordinating concurrent state with Go **goroutines, channels, and mutexes**, and built an interactive **React** dashboard to visualize live cluster state and trigger leader failure on demand.

Chaukidar – Multilingual AI Safety Audit Platform | *FastAPI, LLMs* — GitHub | Live Demo

- Led a team to build **Chaukidar** for the **AMD Developer Hackathon: ACT II (Track 3)**, testing whether LLM safety refusals that hold in English **weaken when the same harmful intent is expressed in South Asian languages** (Urdu, Punjabi, Pashto, Sindhi).
- Designed a controlled evaluation spanning **5 harm categories** across **3 prompt tracks** (English, translation-baseline, native-adapted) over a 180-prompt dataset, producing evidence that **native-language prompts elicit measurably weaker refusals**.
- Built the pipeline end-to-end – a **FastAPI** backend and a **Next.js** frontend – with batched inference on **AMD GPUs via ROCm/vLLM** and a multilingual LLM-judge to avoid the blind spots of English keyword-matching.

Seedha Rasta – Fuel-Optimized Routing Engine | *Django, React, AWS* — GitHub

- Engineered an **A*-based graph routing system** optimizing for estimated fuel cost rather than shortest distance, combining distance-based and traffic-weighted idle consumption in a custom cost function over an OSMnx/NetworkX road-network graph.
- Built the routing backend with **Django + DRF**, cached frequent routing data with **Redis**, and containerized a **three-tier application** using multi-stage Docker builds.
- Deployed on **AWS ECS** behind an Application Load Balancer with RDS, and set up a **GitHub Actions** CI/CD pipeline for automated build, test, and deployment on every push.

Synthron – AI-Powered PC Builder Platform | *Django, PostgreSQL, Docker, LLMs* — GitHub

- Built an AI-powered recommendation platform using **Django REST Framework, PostgreSQL, and local LLM-based intent parsing** (LLaMA 3.2 via Ollama) to convert natural-language requirements into budget- and region-aware hardware builds.
- Designed a modular multi-agent architecture – a **CentralAgent** orchestrating specialized CPU, GPU, RAM, storage, and motherboard agents – separating component-level reasoning from build-level compatibility checking.
- Implemented real-time price/availability scraping with **Playwright** across multiple e-commerce sources, and integrated **Redis caching, JWT auth, and Docker Compose** across a 5-service stack for reproducible local development.

Technical Skills

Languages: Go (Golang), Python, C++, Java, JavaScript, TypeScript
Frontend & Backend: React, Next.js, Express.js, Django
Databases: MongoDB, PostgreSQL, Redis, Firebase
DevOps & Cloud: Docker, AWS, GitHub Actions, CI/CD